

Ropivacaine: Drug information - UpToDate

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Ropivacaine: Drug information

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For additional information see "Ropivacaine: Patient drug information" and "Ropivacaine: Pediatric drug information"

For abbreviations, symbols, and age group definitions show table

Brand Names: US

- Naropin

Brand Names: Canada

- Naropin

Pharmacologic Category

- Local Anesthetic

Dosing: Adult

Dose varies with procedure, onset and depth of anesthesia desired, vascularity of tissues, duration of anesthesia, and condition of patient. A test dose (eg, 3 to 5 mL) of short-acting local anesthetic containing epinephrine should be administered prior to epidural anesthesia or induction of complete block with ropivacaine. Incremental ropivacaine dosing is recommended. Adults:

Anesthesia, surgical

Anesthesia, surgical:

Lumbar epidural block for surgery:

15 to 30 **mL** of 0.5% solution

15 to 25 **mL** of 0.75% solution

15 to 20 **mL** of 1% solution

Lumbar epidural block for cesarean delivery:

20 to 30 **mL** dose of 0.5% solution

15 to 20 **mL** dose of 0.75% solution

Thoracic epidural block:

5 to 15 **mL** dose of 0.5% solution

5 to 15 **mL** dose of 0.75% solution

Major nerve block:

35 to 50 **mL** dose of 0.5% solution

10 to 40 **mL** dose of 0.75% solution

Field block: 1 to 40 **mL** dose of 0.5% solution

Labor pain management

Labor pain management: Lumbar epidural: Initial: 10 to 20 **mL** 0.2% solution; continuous infusion dose: 6 to 14 **mL**/hour of 0.2% solution with intermittent bolus injections of up to 10 to 15 **mL**/hour of 0.2% solution.

Postoperative pain management

Postoperative pain management:

Peripheral nerve block: Continuous infusion dose: 5 to 10 **mL**/hour of 0.2% solution (Ref)

Lumbar or thoracic epidural: Continuous infusion dose: 6 to 14 **mL**/hour of 0.2% solution

Infiltration/minor nerve block:

1 to 100 **mL** dose of 0.2% solution

1 to 40 **mL** dose of 0.5% solution

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Adult

There are no dosage adjustments provided in the manufacturer's labeling. However, ropivacaine and its metabolites are renally excreted, and the risk of toxic reactions may be greater.

Dosing: Liver Impairment: Adult

There are no dosage adjustments provided in the manufacturer's labeling. Use with caution; ropivacaine undergoes hepatic metabolism and patients may be at a greater risk for developing toxic drug levels.

Dosing: Older Adult

Refer to adult dosing. Use with caution; initial dose reductions may be necessary.

Dosing: Pediatric

(For additional information see "Ropivacaine: Pediatric drug information")

Note: Dose varies with procedure, depth of anesthesia, vascularity of tissues, duration of anesthesia, and condition of patient. Consider incremental administration with negative aspiration prior to each injection; however, absence of blood in the syringe does not guarantee that intravascular injection has been avoided (Ref). Should only be administered under the supervision of a qualified physician experienced in the use of anesthetics. Dosing units are variable (**mL/kg**, **mg/kg**); use extra precaution to ensure accuracy and minimize potential toxicity.

Central nerve blocks/anesthesia

Central nerve blocks/anesthesia (eg, caudal, lumbar, thoracic): Limited data available:

Caudal block: Infants and Children: Caudal injection: 0.2% (2 mg/mL) solution: 0.5 to 1 mL/kg; some experts suggest a maximum volume of 25 mL (Ref).

Epidural block (eg, lumbar, thoracic): Infants, Children, and Adolescents: Epidural injection 0.2% (2 mg/mL): 0.7 mL/kg. **Note:** For infants (particularly young infants) if repeat injections necessary, a decreased dose may be necessary to prevent drug accumulation. Some experts suggest if at least 45 minutes since initial dose, reduce dose to $\frac{1}{3}$ of the initial or if at least 90 minutes since initial

dose, then reduce dose to ½ of the initial. If additional doses are necessary, doses should be reduced to ½ of the previous dose (Ref).

Epidural, continuous infusion:

Infants <3 months: Epidural injection: Bolus: Usual concentration 0.2% (2 mg/mL) solution: Reported range: 0.5 to 1 mL/kg. In some cases, a more dilute solution (eg, 0.1% [1 mg/mL]) may be required to ensure adequate volume and minimize toxicity. Follow with continuous epidural infusion of 0.2% (2 mg/mL) solution at 0.2 **mg/kg**/hour (Ref).

Infants ≥3 months, Children, and Adolescents: Epidural injection: 0.2% (2 mg/mL) solution: Bolus: Reported range: 0.5 to 1 mL/kg administered over several minutes (eg, 3 to 5 minutes) followed by continuous epidural infusion at 0.2 to 0.5 **mg/kg**/hour; in trials, 0.4 **mg/kg**/hour was frequently reported; some suggest a maximum infusion rate of 0.2 **mg/kg**/hour for infants <6 months of age although higher rates have been reported; in adults, usual rates are 6 to 14 mL/hour of 0.2% solution (Ref).

Peripheral nerve blocks/local anesthesia

Peripheral nerve blocks/local anesthesia: Limited data available: **Note:** Dose varies with location of block (ie, procedure), depth of anesthesia, vascularity of tissues, duration of anesthesia, and patient parameters (eg, age, weight, condition).

Single injection: The volume of dose (mL/kg) and concentration of solution are site-specific based upon anatomy and variable among patients and procedure (Ref).

Infants ≥6 months, Children, and Adolescents: Suggested or reported dose volumes presented; not to exceed a suggested maximum dose of 3 **mg/kg**/dose based on lean body mass; dosing based on extrapolation from adult experience; additional data necessary to more clearly define the pediatric upper dose limit. **Note:** For infants <6 months of age, dose reductions (eg, by 30%) have been suggested by experts (Ref).

Commonly accepted doses (Ref):

Head and neck blocks: 0.05 mL/kg.

Maxillary nerve: 0.15 mL/kg (Ref).

Upper extremity blocks:

Brachial plexus: 0.2 to 0.3 mL/kg.

Digital nerve: ≤0.2% (2 mg/mL) solution: 0.05 mL/kg.

Truncal blocks:

Transversus abdominis plane: 0.2 to 0.5 mL/kg (Ref).

Ilioinguinal nerve: 0.075 mL/kg.

Rectus sheath: 0.1 to 0.2 mL/kg.

Lower extremity blocks:

Femoral nerve: 0.2% to 0.5% (2 to 5 mg/mL) solution: 0.2 to 0.4 mL/kg (Ref).

Sciatic nerve: 0.2 to 0.3 mL/kg.

Continuous peripheral nerve block infusion (CPNB): Dosing based on trial experience (Ref):

Infants ≥6 months, Children, and Adolescents:

Initial bolus: 0.2% (2 mg/mL) solution: Dose dependent on nerve catheter location: Commonly reported dose range: ~0.5 to 1.32 **mg/kg**; some patients and/or catheter/nerve sites may require a higher bolus dose; a trial evaluating 403 pediatric catheter placements reported an overall median bolus for all catheter sites of 1.32 **mg/kg** (interquartile range [IQR]: 0.82 to 1.95 **mg/kg**); another trial reporting on 339 pediatric catheter placements reported a mean initial bolus dose of 0.49 to 0.98 **mg/kg**.

Continuous peripheral nerve block infusion: 0.2% (2 mg/mL) solution: Reported mean rate range: 0.11 to 0.25 **mg/kg**/hour (most experience reported at 0.2 to 0.25 **mg/kg**/hour); reported IQR range from 403 pediatric catheter placements: 0.12 to 0.33 **mg/kg**/hour (depending upon catheter site) with a usual duration <72 hours; the majority of experience is postoperative pain management following orthopedic procedures including administration via On-Q pump or Ambu Smart-Infuser.

Spinal anesthesia

Spinal anesthesia: Limited data available: Infant, Children, and Adolescents ≤ 17 years: Intrathecal (via LP site; L 3-5): Preservative-free 0.5% (5 mg/mL) isobaric solution: 0.5 **mg/kg**; maximum dose: 20 mg/dose (Ref).

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Pediatric

There are no dosage adjustments provided in the manufacturer's labeling. However, ropivacaine and its metabolites are renally excreted, and the risk of toxic reactions may be greater.

Dosing: Liver Impairment: Pediatric

There are no dosage adjustments provided in the manufacturer's labeling. Use with caution; ropivacaine undergoes hepatic metabolism and patients may be at a greater risk for developing toxic drug levels.

Adverse Reactions

The following adverse drug reactions and incidences are derived from product labeling unless otherwise specified.

>10%:

Cardiovascular: Bradycardia (6% to 20%), hypotension (32% to 69%)

Gastrointestinal: Nausea (13% to 25%), vomiting (7% to 12%)

Neuromuscular & skeletal: Back pain (4% to 16%)

1% to 10%:

Cardiovascular: Chest pain (1% to 5%), hypertension (1% to 5%), tachycardia (1% to 5%)

Dermatologic: Pruritus (1% to 5%)

Endocrine & metabolic: Hypokalemia (1% to 5%)

Genitourinary: Disorder of breast milk secretion (1%), oliguria (1% to 5%), poor progression of labor (1%), urinary retention (1% to 5%), urinary tract infection (1% to 5%)

Hematologic & oncologic: Anemia (6%)

Nervous system: Anxiety (1%), chills ($\leq 3\%$), dizziness (3%), headache (5% to 8%), hypoesthesia (2%), pain (4% to 8%), paresthesia (2% to 6%), rigors ($\leq 3\%$)

Neuromuscular & skeletal: Muscle cramps (1% to 5%)

Respiratory: Dyspnea (1% to 5%), rhinitis (1%)

Miscellaneous: Fever (2% to 9%)

<1%:

Cardiovascular: Acute myocardial infarction, atrial fibrillation, cardiac arrhythmia, deep vein thrombosis, ECG abnormality, extrasystoles, orthostatic hypotension, phlebitis, pulmonary embolism, ST segment changes on ECG, syncope

Dermatologic: Skin rash, urticaria

Endocrine & metabolic: Hypomagnesemia

Gastrointestinal: Fecal incontinence, tenesmus

Genitourinary: Urinary incontinence, urination disorder, uterine atony

Hepatic: Jaundice

Local: Pain at injection site

Nervous system: Agitation, amnesia, asthenia, coma, confusion, drowsiness, emotional lability, hallucination, Horner syndrome, hypothermia, hypotonia, insomnia, malaise, nervousness, neuropathy, nightmares, paresis, seizure, stupor, tremor, vertigo

Neuromuscular & skeletal: Dyskinesia, hypokinesia, myalgia

Ophthalmic: Blepharoptosis, visual disturbance

Otic: Auditory disturbance, tinnitus

Respiratory: Bronchospasm, cough

Miscellaneous: Accidental injury

Contraindications

Hypersensitivity to ropivacaine, amide-type local anesthetics (eg, bupivacaine, mepivacaine, lidocaine), or any component of the formulation

Canadian labeling: Additional contraindications (not in US labeling): Intravenous regional anesthesia (Bier block); obstetric paracervical block anesthesia

Warnings/Precautions

Concerns related to adverse effects:

- **CNS toxicity:** Careful and constant monitoring of the patient's state of consciousness should be done following each local anesthetic injection; at such times, restlessness, anxiety, tinnitus, dizziness, blurred vision, tremors, depression, or drowsiness may be early warning signs of CNS toxicity. Treatment is primarily symptomatic and supportive.
- **Intra-articular infusion related chondrolysis:** Continuous intra-articular infusion of local anesthetics after arthroscopic or other surgical procedures is **not** an approved use; chondrolysis (primarily in the shoulder joint) has occurred following infusion, with some cases requiring arthroplasty or shoulder replacement.
- **Methemoglobinemia:** Has been reported with local anesthetics; clinically significant methemoglobinemia requires immediate treatment along with discontinuation of the anesthetic and other oxidizing agents. Onset may be immediate or delayed (hours) after anesthetic exposure. Patients with glucose-6-phosphate dehydrogenase deficiency, congenital or idiopathic methemoglobinemia, cardiac or pulmonary compromise, exposure to oxidizing agents or their metabolites, or infants <6 months of age are more susceptible and should be closely monitored for signs and symptoms of methemoglobinemia (eg, cyanosis, headache, rapid pulse, shortness of breath, lightheadedness, fatigue).
- **Respiratory arrest:** Local anesthetics have been associated with rare occurrences of sudden respiratory arrest.
- **Seizures:** Convulsions due to systemic toxicity leading to cardiac arrest have also been reported, presumably following unintentional intravascular injection.

Disease-related concerns:

- **Cardiovascular disease:** Use with caution in patients with hypotension, hypovolemia, heart block, or cardiovascular disease; may be at greater risk for toxicity.
- **Hepatic impairment:** Use with caution in patients with hepatic impairment; may be at greater risk for toxicity.

- Neurological disorders: Use with caution in patients with neurological disorders; may be at greater risk for toxicity.
- Porphyria: Use with caution in patients with acute porphyria; consider use of alternative agents.
- Psychiatric disorders: Use with caution in patients with psychiatric disorders; may be at greater risk for toxicity.
- Renal impairment: Use with caution in patients with severe renal impairment; may be at greater risk for toxicity.

Special populations:

- Acutely ill patients: Use with caution in acutely ill; may be at greater risk for toxicity.
- Debilitated patients: Use with caution in debilitated patient; may be at greater risk for toxicity.
- Older adult: Use with caution in the elderly: may be at greater risk for toxicity. Cardiovascular adverse events (bradycardia, hypotension) may be age-related (more common in patients >61 years of age).

Other warnings/precautions:

- Administration: Intravascular injections should be avoided; aspiration should be performed prior to administration; the needle must be repositioned until no return of blood can be elicited by aspiration; however, absence of blood in the syringe does not guarantee that intravascular injection has been avoided.
- Trained personnel: Clinicians using local anesthetic agents should be well trained in diagnosis and management of emergencies that may arise from the use of these agents. Resuscitative equipment, oxygen, and other resuscitative drugs should be available for immediate use.

Dosage Forms: US

Excipient information presented when available (limited, particularly for generics); consult specific product labeling.

Solution, Injection, as hydrochloride:

Generic: 2 mg/mL (100 mL)

Solution, Injection, as hydrochloride [preservative free]:

Naropin: 2 mg/mL (10 mL, 20 mL, 100 mL, 200 mL); 5 mg/mL (20 mL, 30 mL, 100 mL, 200 mL); 7.5 mg/mL (20 mL); 10 mg/mL (10 mL, 20 mL)

Generic: 2 mg/mL (10 mL, 20 mL, 100 mL, 200 mL); 5 mg/mL (20 mL, 30 mL, 100 mL, 200 mL); 7.5 mg/mL (20 mL); 10 mg/mL (10 mL, 20 mL)

Generic Equivalent Available: US

Yes

Pricing: US

Solution (Naropin Injection)

2 mg/mL (per mL): \$0.24

5 mg/mL (per mL): \$0.20

7.5 mg/mL (per mL): \$1.13

10 mg/mL (per mL): \$1.36

Solution (ROPivacaine HCl Injection)

2 mg/mL (per mL): \$0.18 - \$0.72

5 mg/mL (per mL): \$0.16 - \$0.73

7.5 mg/mL (per mL): \$1.20 - \$1.62

10 mg/mL (per mL): \$0.48 - \$1.67

Disclaimer: A representative AWP (Average Wholesale Price) price or price range is provided as reference price only. A range is provided when more than one manufacturer's AWP price is available and uses the low and high price reported by the manufacturers to determine the range. The pricing data should be used for benchmarking purposes only, and as such should not be used alone to set or adjudicate any prices for reimbursement or purchasing functions or considered to be an exact price for a single product and/or manufacturer. Medi-Span expressly disclaims all warranties of any kind or nature, whether express or implied, and assumes no liability with respect to accuracy of price or price range data published in its solutions. In no event shall Medi-Span be liable for special, indirect, incidental, or consequential damages arising from use of price or price range data. Pricing data is updated monthly.

Dosage Forms: Canada

Excipient information presented when available (limited, particularly for generics); consult specific product labeling.

Solution, Injection, as hydrochloride:

Naropin: 2 mg/mL (100 mL, 200 mL); 5 mg/mL (20 mL); 10 mg/mL (10 mL, 20 mL)

Generic: 2 mg/mL (10 mL, 20 mL, 100 mL, 200 mL); 5 mg/mL (10 mL); 10 mg/mL (10 mL, 20 mL)

Administration: Adult

Administered via local infiltration, epidural block and epidural infusion, or intermittent bolus. Avoid rapid administration of large volumes of ropivacaine; use fractional (incremental) doses with the lowest effective dose and concentration required to produce the desired result. Prior to epidural anesthesia or induction of complete block, a test dose (eg, 3 to 5 mL) of short-acting local anesthetic with epinephrine should be administered. The On-Q infusion pump is used to slowly administer local anesthetics (eg, bupivacaine, lidocaine, ropivacaine) to or around surgical wound sites and/or in close proximity to nerves for pre- or postoperative regional anesthesia. When infused directly into the shoulder, destruction of articular cartilage (chondrolysis) has occurred. On-Q pumps should never be placed directly into any joint (Ref).

Administration: Pediatric

Parenteral: Administered via local infiltration, epidural block and epidural infusion, or intermittent bolus; if further dilution necessary, ensure appropriate diluent (eg, preservative-free). Prior to ropivacaine administration as epidural anesthesia or induction of complete block, a test dose of short acting local anesthetic with epinephrine should be administered to verify avoidance of intravascular site. Incremental ropivacaine dosing is recommended. Do not administer large volume of anesthetic rapidly. Various infusion pumps are on market and used to slowly administer local anesthetics (eg, bupivacaine, lidocaine, ropivacaine) to or around surgical wound sites and/or in close proximity to nerves for pre- or postoperative regional anesthesia; in most pediatric trials, an On-Q pump was used. When infused directly into the shoulder, destruction of articular cartilage (chondrolysis) has occurred. On-Q pumps should never be placed directly into any joint (Ref).

Storage/Stability

Store at 20°C to 25°C (68°F to 77°F); excursions permitted to 15°C between 30°C (59°F to 86°F). Infusions should be discarded after 24 hours.

Use: Labeled Indications

Acute pain management: For acute pain management administered as an epidural continuous infusion, intermittent bolus (eg, postoperative or labor), or local infiltration.

Surgical anesthesia: For the production of local or regional anesthesia for surgery administered as an epidural block, including cesarean delivery, major nerve block, or local infiltration.

Medication Safety Issues

Sound-alike/look-alike issues:

ROPivacaine may be confused with BUPivacaine, rOPINIRole

Infusion bottles of Naropin (ropivacaine) and Ofirmev (acetaminophen) look similar. Potentially fatal mix-ups have been reported in which a glass bottle of Naropin was mistaken for Ofirmev in perioperative areas.

High alert medication:

The Institute for Safe Medication Practices (ISMP) includes this medication among its list of drug classes (epidural and intrathecal medications) which have a heightened risk of causing significant patient harm when used in error (ISMP List of High-Alert Medications in Acute Care Settings ©ISMP 2024).

Metabolism/Transport Effects

Substrate of CYP1A2 (Major with inhibitors), CYP1A2 (Minor with inducers), CYP2B6 (Minor), CYP2D6 (Minor), CYP3A4 (Minor); **Note:** Assignment of Major/Minor substrate status based on clinically relevant drug interaction potential

Drug Interactions

(For additional information: Launch drug interactions program)

Note: Interacting drugs may **not be individually listed below** if they are part of a group interaction (eg, individual drugs within “CYP3A4 Inducers [Strong]” are NOT listed). For a complete list of drug interactions by individual drug name and detailed management recommendations, use the drug interactions program by clicking on the “Launch drug interactions program” link above.

Alfuzosin: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Amifostine: Blood Pressure Lowering Agents may increase hypotensive effects of Amifostine. Management: When used at chemotherapy doses, hold blood pressure lowering medications for 24 hours before amifostine administration. If blood pressure lowering therapy cannot be held, do not administer amifostine. Use caution with radiotherapy doses of amifostine. *Risk D: Consider Therapy Modification*

Amisulpride (Oral): May increase hypotensive effects of Hypotension-Associated Agents. *Risk C: Monitor*

Antiarrhythmic Agents (Class III): May increase arrhythmogenic effects of ROPivacaine. *Risk C: Monitor*

Antipsychotic Agents (Second Generation [Atypical]): Blood Pressure Lowering Agents may increase hypotensive effects of Antipsychotic Agents (Second Generation [Atypical]). *Risk C: Monitor*

Arginine: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Articaine: May increase adverse/toxic effects of Local Anesthetics. *Risk C: Monitor*

Barbiturates: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Benperidol: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Blood Pressure Lowering Agents: May increase hypotensive effects of Hypotension-Associated Agents. *Risk C: Monitor*

Brimonidine (Topical): May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Bromperidol: May decrease hypotensive effects of Blood Pressure Lowering Agents. Blood Pressure Lowering Agents may increase hypotensive effects of Bromperidol. *Risk X: Avoid*

BUPIvacaine (Liposomal): Local Anesthetics may increase adverse/toxic effects of BUPivacaine (Liposomal). Management: Liposomal bupivacaine should not be administered with local anesthetics, but may be administered 20 minutes or more after lidocaine. Avoid all local anesthetics within 96 hours after administration of liposomal bupivacaine. *Risk X: Avoid*

BUPivacaine: Local Anesthetics may increase adverse/toxic effects of BUPivacaine. Management: Avoid using any additional local anesthetics within 96 hours after insertion of the bupivacaine implant (Xaracoll) or bupivacaine and meloxicam periarticular solution (Zynrelef) or within 168 hours after subacromial infiltration (Posimir brand). *Risk C: Monitor*

CYP1A2 Inhibitors (Moderate): May increase serum concentrations of ROPivacaine. *Risk C: Monitor*

CYP1A2 Inhibitors (Strong): May increase serum concentrations of ROPivacaine. *Risk C: Monitor*

Diazoxide (Immediate Release): May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Diazoxide Choline (Extended Release): May increase serum concentrations of CYP1A2 Substrates (High risk with Inhibitors). Management: Consider alternatives when possible, as product labeling states that coadministration of diazoxide choline and CYP1A2 substrates is not recommended. If combined, monitor for increased effects of the CYP1A2 substrate. *Risk D: Consider Therapy Modification*

DULoxetine: Blood Pressure Lowering Agents may increase hypotensive effects of DULoxetine. *Risk C: Monitor*

Herbal Products with Blood Pressure Lowering Effects: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Hyaluronidase: May increase adverse/toxic effects of Local Anesthetics. *Risk C: Monitor*

Hypotension-Associated Agents: Blood Pressure Lowering Agents may increase hypotensive effects of Hypotension-Associated Agents. *Risk C: Monitor*

Iloperidone: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Levodopa-Foslevodopa: Blood Pressure Lowering Agents may increase hypotensive effects of Levodopa-Foslevodopa. *Risk C: Monitor*

Local Anesthetics: ROPivacaine may increase adverse/toxic effects of Local Anesthetics. *Risk C: Monitor*

Lormetazepam: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Metergoline: May decrease antihypertensive effects of Blood Pressure Lowering Agents. Blood Pressure Lowering Agents may increase orthostatic hypotensive effects of Metergoline. *Risk C: Monitor*

Methemoglobinemia Associated Agents: May increase adverse/toxic effects of Local Anesthetics. Specifically, the risk for methemoglobinemia may be increased. *Risk C: Monitor*

Milsaperidone: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Molsidomine: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Naftopidil: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Neuromuscular-Blocking Agents: Local Anesthetics may increase neuromuscular-blocking effects of Neuromuscular-Blocking Agents. *Risk C: Monitor*

Nicergoline: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Nicorandil: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Nitroprusside: Blood Pressure Lowering Agents may increase hypotensive effects of Nitroprusside. *Risk C: Monitor*

Obinutuzumab: May increase hypotensive effects of Blood Pressure Lowering Agents. Management: Consider temporarily withholding blood pressure lowering medications beginning 12 hours prior to

obinutuzumab infusion and continuing until 1 hour after the end of the infusion. *Risk D: Consider Therapy Modification*

Pentoxifylline: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Pholcodine: Blood Pressure Lowering Agents may increase hypotensive effects of Pholcodine. *Risk C: Monitor*

Phosphodiesterase 5 Inhibitors: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Primaquine: May increase serum concentrations of CYP1A2 Substrates (High risk with Inhibitors). *Risk C: Monitor*

Prostacyclin Analogues: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

PyRIDostigmine: May increase or decrease neuromuscular-blocking effects of Local Anesthetics. *Risk C: Monitor*

Rasagiline: Blood Pressure Lowering Agents may increase hypotensive effects of Rasagiline. *Risk C: Monitor*

Selegiline: Blood Pressure Lowering Agents may increase hypotensive effects of Selegiline. *Risk C: Monitor*

Silodosin: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Technetium Tc 99m Tilmanocept: Coadministration of Local Anesthetics and Technetium Tc 99m Tilmanocept may alter diagnostic results. Management: Avoid mixing and simultaneously co-injecting technetium Tc 99m tilmanocept with local anesthetics. This interaction does not appear to apply to other uses of these agents in combination. *Risk C: Monitor*

Tolcapone: Blood Pressure Lowering Agents may increase hypotensive effects of Tolcapone. *Risk C: Monitor*

Pregnancy Considerations

Ropivacaine crosses the placenta following epidural administration (Matsota 2009; Simpson 2005).

Ropivacaine may potentially cause varying degrees of maternal, fetal, and neonatal toxicity involving the CNS, peripheral vascular tone, and cardiac function depending on dose and administration technique.

Ropivacaine is approved for use in obstetric analgesia/anesthesia (ACOG 2019; Reschke 2020).

Breastfeeding Considerations

Ropivacaine is present in breast milk.

Data related to the presence of ropivacaine in breast milk are available from a study of 25 patients administered ropivacaine 0.15% combined with fentanyl for patient-controlled epidural analgesia following elective cesarean delivery. The mean ($\pm$ SD) accumulated doses of ropivacaine following epidural administration were 187.7 ± 53.9 mg and 248.3 ± 61.8 mg, at 18 and 24 hours, respectively. Corresponding maternal plasma concentrations were 979 ± 263 ng/mL (18 hours), and $1,282 \pm 261$ ng/mL (24 hours). Breast milk concentrations of ropivacaine were 246 ± 98 ng/mL at 18 hours and 301 ± 122 ng/mL at 24 hours. Adverse events were not observed in the patients or breastfed neonates (Matsota 2009).

According to the manufacturer, the decision to breastfeed during therapy should consider the risk of infant exposure, the benefits of breastfeeding to the infant, and the benefits of treatment to the mother; however, local anesthetics are considered compatible with breastfeeding (ABM [Martin 2018]; ABM [Reece-Stremtan 2017]; Mitchell 2020).

Monitoring Parameters

Heart rate, blood pressure, ECG monitoring (if used with antiarrhythmics)

Mechanism of Action

Blocks both the initiation and conduction of nerve impulses by decreasing the neuronal membrane's permeability to sodium ions, which results in inhibition of depolarization with resultant blockade of conduction

Pharmacokinetics (Adult Data Unless Noted)

Onset of action: Anesthesia (route dependent): 3 to 15 minutes

Duration (dose and route dependent): 3 to 15 hours

Protein binding: 94%, primarily to alpha1-acid glycoprotein

Distribution: V_d :

Children: Epidural infusion: 2.1 to 4.2 L/kg (Hansen 2000)

Adults: Intravascular infusion: 41 ± 7 L

Metabolism: Hepatic, via CYP1A2 to metabolites

Half-life elimination:

Children: Epidural: Terminal phase: 4.9 hours (range: 3 to 6.7 hours) (Hansen 2000)

Adults: Epidural: 5 to 7 hours; IV: Terminal: 111 ± 62 minutes (Lee, 1989)

Time to peak, serum: Dose and route dependent: Caudal:

Infants: Median: 60 minutes (range: 15 to 90 minutes) (Wulf 2000)

Children: Mean: 60 minutes (range: 12 to 249 minutes (Lonnqvist 2000)

Excretion: Urine (86%; 1% as unchanged drug)

Patient Counseling Points

(For additional information see "Ropivacaine: Patient drug information")

What is this drug used for?

- It is used to numb an area before a procedure.
- It is used to manage pain.

All drugs may cause side effects. However, many people have no side effects or only have minor side effects. Call your doctor or get medical help if any of these side effects or any other side effects bother you or do not go away:

- Headache
- Upset stomach or throwing up
- Back pain

WARNING/CAUTION: Even though it may be rare, some people may have very bad and sometimes deadly side effects when taking a drug. Tell your doctor or get medical help right away if you have any of the following signs or symptoms that may be related to a very bad side effect:

For all uses of this drug:

- Too much acid in the blood (acidosis) like confusion; fast breathing; fast heartbeat; a heartbeat that does not feel normal; very bad stomach pain, upset stomach, or throwing up; feeling very sleepy; shortness of breath; or feeling very tired or weak
- Methemoglobinemia like a blue or gray color of the lips, nails, or skin; a heartbeat that does not feel normal; seizures; severe dizziness or passing out; severe headache; feeling very sleepy; feeling tired or weak; or shortness of breath

- Feeling very tired or weak
- Chest pain
- Fast, slow, or abnormal heartbeat
- A burning, numbness, or tingling feeling that is not normal
- Change in balance
- Restlessness
- Anxiety
- Change in speech
- Numbness or tingling in the mouth
- Metallic taste
- Ringing in ears
- Dizziness or passing out
- Feeling lightheaded, sleepy, confused, or having blurred eyesight
- Shakiness
- Twitching
- Feeling hot or cold
- Very nervous and excitable
- Depression
- Seizures
- Trouble breathing, slow breathing, or shallow breathing
- Sneezing
- Sweating a lot
- Signs of an allergic reaction, like rash; hives; itching; red, swollen, blistered, or peeling skin with or without fever; wheezing; tightness in the chest or throat; trouble breathing, swallowing, or talking; unusual hoarseness; or swelling of the mouth, face, lips, tongue, or throat.

Epidural:

- Trouble passing urine
- Loss of bladder or bowel control
- Not able to get or keep an erection
- Long-lasting burning, numbness, tingling, or paralysis in the lower half of the body
- Backache
- Fever or chills
- Stiff neck
- If bright lights bother your eyes

Note: This is not a comprehensive list of all side effects. Talk to your doctor if you have questions.

Consumer Information Use and Disclaimer: This information should not be used to decide whether or not to take this medicine or any other medicine. Only the healthcare provider has the knowledge and training to decide which medicines are right for a specific patient. This information does not endorse any medicine as safe, effective, or approved for treating any patient or health condition. This is only a limited summary of general information about the medicine's uses from the

patient education leaflet and is not intended to be comprehensive. This limited summary does NOT include all information available about the possible uses, directions, warnings, precautions, interactions, adverse effects, or risks that may apply to this medicine. This information is not intended to provide medical advice, diagnosis or treatment and does not replace information you receive from the healthcare provider. For a more detailed summary of information about the risks and benefits of using this medicine, please speak with your healthcare provider and review the entire patient education leaflet.

Brand Names: International

International Brand Names by Country

For country code abbreviations (show table)

- (AT) Austria: Naropin | Ropinaest | Ropivacain Actavis | Ropivacain Chiesi | Ropivacain sintetica;
- (AU) Australia: Naropin | Ropibam | Ropivacaine aft | Ropivacaine kabi | Ropivacaine Sandoz;
- (BE) Belgium: Naropin | Ropivacain Sandoz | Ropivacaine Fresenius Kabi;
- (BR) Brazil: Naropin | Ropi | Ropi bolsa;
- (CH) Switzerland: Ropivacain sintetica;
- (CN) China: Ropivacaine hydrochloride and sodium chloride;
- (CZ) Czech Republic: Naropin;
- (DE) Germany: Naropin | Ropivacain Actavis | Ropivacain altan | Ropivacain bioq | Ropivacain cipla | Ropivacain deltamedica | Ropivacain hcl noridem | Ropivacain Hexal | Ropivacain sintetica | Ropivacainhydrochlorid GRY;
- (EE) Estonia: Naropin | Ropivacaine;
- (ES) Spain: Naropin polybag | Ropivacaina g.e.s | Ropivacaina inibsa | Ropivacaina Kabi;
- (FI) Finland: Naropin;
- (FR) France: Naropeine | Ropivacaine altan | Ropivacaine b.braun | Ropivacaine kabi | Ropivacaine Mylan | Ropivacaine noridem | Ropivacaine teva;
- (GB) United Kingdom: Naropin | Ropivacaine;
- (GR) Greece: Naropeine | Ropivacaine/kabi | Ropivacaine/teva | Zitamin;
- (HU) Hungary: Naropin;
- (ID) Indonesia: Naropin;
- (IE) Ireland: Naropin;
- (IT) Italy: Naropina | Ropivacaina | Ropivacaina cloridrato altan pharma | Ropivacaina cloridrato s.a.l.f. | Ropivacaina galenica senese | Ropivacaina molteni | Ropivacaina Sandoz;
- (JP) Japan: Anapeine;
- (KR) Korea, Republic of: Naropin;
- (LT) Lithuania: Ropivacaine b.braun;
- (LV) Latvia: Naropin;
- (MY) Malaysia: Naropin | Ropivacain sintetica;
- (NL) Netherlands: Naropin | Ropivacaine;
- (NO) Norway: Naropin;
- (NZ) New Zealand: Naropin | Ropibam | Ropivacaine kabi;
- (PL) Poland: Ropivacaine bioq;
- (PR) Puerto Rico: Naropin | Ropivacaine HCL;
- (PT) Portugal: Naropeine | Ropivacaina | Ropivacaina actavis;
- (QA) Qatar: Naropin;
- (RU) Russian Federation: Naropin | Ropivabin;
- (SE) Sweden: Narop | Ropivacain fresenius kabi | Ropivakain Actavis;
- (SG) Singapore: Naropin;
- (SI) Slovenia: Naropin;
- (TH) Thailand: Naropin;
- (TN) Tunisia: Ropivacaine Infomed | Ropivacaine kabi;
- (TR) Turkey: Naropin;
- (UA) Ukraine: Naropin | Ropilong;

- (VE) Venezuela, Bolivarian Republic of: Naropin;
- (ZA) South Africa: Adco ropivacaine | Naropin

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